# UPVAAS-Twin

A festival- and fasting-aware metabolic digital twin
for Indians with type 2 diabetes

upvaas · Uncertainty-aware Physiology-informed Virtual Avatar for Adverse-event Surveillance

Happiest Health · Reimagining and Reforming Healthcare in India Summit 2026 · Digital Twin Challenge

Team <Team Name> · <College / Incubator>

# A danger that is already on the calendar

Scheduled, common, unmonitored - and therefore predictable

### The scale

101 million adults with diabetes
136 million with prediabetes
(ICMR-INDIAB, Lancet D&E 2023)

### The drug

57% take a sulfonylurea,
which keeps releasing insulin
even when no food arrives

### The behaviour

49% fast for religion
4% check glucose while fasting
1.35% ask a doctor first

**Navratri · Ekadashi · Karva Chauth (nirjala) · Ramadan roza**

A morning glimepiride plus a 13-hour fast is a predictable hypoglycaemia. The fast-breaking thali that follows is a predictable spike. Existing twins treat these days as noise.

# Static data sets the prior. Live data updates the posterior.

The calendar tells the twin what is coming - days in advance

### 1 · EHR prior

labs, meds, eGFR,
CYP2C9 and TCF7L2
-> personal physiology
before any device data

### 2 · Mechanistic twin

Bergman + gut + sulfonylurea
PK/PD + fasting clock
+ blunted counter-regulation
-> explains WHY, not just what

### 3 · Particle filter

64 candidate physiologies
re-weighted by every reading
-> a twin that follows the
patient, not a classifier

### Vrat Rehearsal - the part that makes it a twin

"If she fasts tomorrow and takes her usual glimepiride, what happens?"
36-hour counterfactual replay on the posterior particles:
usual dose · halved · skipped · moved to the fast-breaking meal
low-GI swap · a walk after iftar · garba at night

### Into the health system

ASHA 'Vrat Saathi' recall list, ranked, in the patient's language
eSanjeevani teleconsult before the fast, not an ambulance after
FHIR R4 RiskAssessment + ABDM-shaped consent artefact
Runs on a laptop CPU, in a Tier-3 clinic, with no CGM

# The result that matters: it works without a CGM

Most Indians with type 2 diabetes will never wear one

Tier B · two fingersticks a day + a basic smartwatch · no CGM

**0.08 → 0.13**

hypoglycaemia AUPRC

1.6x, adding the twin

**5.4 → 2.7**

false alarms per patient-day

the twin filters, not just ranks

**26%**

precision of a 20-name worklist

against a 0.9% base rate

**Of the 20 patients the twin flags each morning, roughly 5 really do have an event within two hours.**

### Forecasting (tier A)

twin RMSE 27 / 33 / 38 mg/dL at 30 / 60 / 120 min
persistence  29 / 44 / 55 mg/dL
Clarke error grid zone A+B: 97.2%

### How it was scored

200 synthetic patients x 28 days
5-fold patient-grouped cross-validation
every patient scored by a model that never saw them
patient-clustered bootstrap CIs on every headline number

# Graceful degradation: the same twin, three levels of device

We publish the whole ladder, so a low-resource clinic knows exactly what it is getting

| sensor tier | hypo AUPRC (95% CI) | precision@20 | event recall | false alarms / patient-day |
|---|---|---|---|---|
| **A · CGM + watch** | 0.220 [0.10–0.34] | 0.36 | 56% | 1.00 |
| **B · 2 fingersticks + watch** | 0.132 [0.05–0.23] | 0.26 | 58% | 2.69 |
| **C · EHR + phone only** | 0.071 [0.02–0.14] | 0.26 | 62% | 6.87 |

**Read this honestly**

The twin helps at every tier, but NOT equally, and the gain grows as the data gets thinner:
tier A 1.3x · tier B 1.6x · tier C 2.1x
Tier B is the one that matters for India: sparse enough that physics has something to add, dense enough to personalise - and it needs no CGM. False alarms still rise sharply as sensors are removed.

# Where it does not work

Including the external validation we ran, and failed

### External validation - and it failed

107 REAL CGM recordings (ShanghaiT2DM), no retraining.
2-hour RMSE 52 mg/dL vs 46 for persistence.
It does not beat 'assume glucose stays put', at any horizon.
We then refit the prior on real patients to fix it.
That failed too - and told us why. ->

### Why it failed - the real finding

Refit on real patients: ratio 1.10 - no change.
Just WIDENING the prior: 1.01 - parity, not better.
The refit's residual spread exploded (Gb0 SD 8 -> 40):
routine EHR fields barely predict physiology in a
REAL cohort. Ours predicted it because we built it that way.

### The baseline that keeps us honest

Four features - planned carbs, HbA1c, time of day.
No glucose. No wearable. No twin.
Hyperglycaemia AUPRC 0.30, against 0.60 for the
full model. We publish it so the reader can see
how much is physiology and how much is timetable.

### Other null results, and limits

The calendar as ML FEATURES adds nothing (0.22 vs 0.22);
its value is the Rehearsal, days earlier.
The drift monitor does not work (AUROC 0.53).
Bands cover 58% above 250 mg/dL, not 90%.
Only 40 of 200 patients ever had a hypo event.

# Uncertainty a clinician can actually rely on

A 90% band that is 90% on average can be 50% exactly where you are looking

| coverage of the 2-hour forecast band | overall | when outcome < 70 | when outcome > 250 | mean band width |
|---|---|---|---|---|
| **Split conformal (one margin)** | 0.90 | 0.95 | 0.47 | 125 mg/dL |
| **Mondrian (margin per glucose band)** | 0.90 | 0.91 | 0.58 | 110 mg/dL |

### Why this slide exists

Split conformal gives a MARGINAL guarantee: 90% averaged over every patient and every minute. Because most calibration points are euglycaemic, the band is far too narrow during exactly the post-meal excursions a clinician is watching. Mondrian conformal calibrates a separate margin per predicted-glucose stratum and restores coverage where it matters. Most published twins report the marginal number only; we report both, and the dashboard draws the Mondrian band.

# Trust, fairness and the regulatory path

Built for ICMR 2023 ethics, DPDP 2023 and the CDSCO draft software guidance

### Audited on the axis that matters here

This project is built on religious fasting, so fasting tradition is the first axis we audit - not an afterthought:
karva_chauth: AUROC 0.980
navratri: AUROC 0.938
none: AUROC 0.981
ramadan: AUROC 0.973
Also audited: sex, urban/rural, age, eGFR, region.

### Governance by design

DPDP 2023 - synthetic data only; consent ID on every record, purpose limitation (CAREMGT), deletion on consent expiry.
ABDM - FHIR R4 bundles, consent artefact via HIE-CM.
CDSCO draft (Oct 2025) - informs, does not drive, a registered practitioner -> likely Class B. Human always in the loop.
ICMR 2023 - model card, subgroup audit, right to refuse AI.

### Deployment path

WHERE NP-NCD screening at Ayushman Arogya Mandirs (~25 million on diabetes treatment under '75 by 25'), private diabetology clinics, CGM programmes.
HOW ASHA app receives the ranked recall list two weeks before Navratri / Ramadan -> home visit or phone call -> eSanjeevani teleconsult -> doctor runs the Vrat Rehearsal -> dose-timing plan in the patient's language.
PROOF (1) retrospective validation on ShanghaiT2DM + CGMacros (2) IRB observational pilot, 100 SU-treated fasters, Navratri 2027 (3) cluster-randomised ASHA recall-list trial; outcome = hypoglycaemia events and ER visits.

# What we are claiming - and what we are not

### We are claiming

A twin that treats India's festival calendar as a planned perturbation, not as noise.
A counterfactual rehearsal a doctor can run BEFORE the fast.
A measured ladder from CGM down to a phone, so the people least likely to own a device are not silently excluded.
Uncertainty, fairness and null results reported, not implied.

### We are not claiming

Clinical validity. The data are synthetic; nothing here has been tested on a real patient.
That the twin helps everywhere - at tier A it does not help precision, and we show that.
That the drift monitor is a safety system yet. It is not.
A medical device. This informs a practitioner; it decides nothing.

**The fast is on the calendar. The hypoglycaemia does not have to be a surprise.**

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